

SQL Learning Journal #12- JOINING DATA

Introduction

Today I started learning how to combine data from multiple tables using the ****UNION**** operator. Combining datasets is an important technique in SQL because information is often stored in separate tables that need to be merged for analysis.

SQL Commands Learned Today

1. UNION ALL and UNION

If two tables have the same number of columns and compatible data types, they can be combined using the ****UNION**** operator.

****UNION ALL**** keeps every row, including duplicate records.

```
SELECT *  
FROM table1_name  
  
UNION ALL  
  
SELECT *  
FROM table2_name  
  
ORDER BY year DESC;
```

The ****UNION**** operator removes duplicate rows automatically.

```
SELECT *  
FROM table1_name  
  
UNION
```

```
SELECT *  
FROM table2_name  
ORDER BY year DESC;
```

2. Important Notes

- Both tables must contain the same number of columns.
- Corresponding columns must have compatible data types and appear in the same order.
- The column names do not have to be identical.
- The result table always uses the column names from the first `SELECT` statement.

What I Learned Today

Today I learned how to combine data from multiple tables using the ****UNION**** operator. I also learned the difference between ****UNION ALL****, which keeps duplicate rows, and ****UNION****, which automatically removes duplicates. Understanding how to combine datasets is an essential step toward working with larger and more complex databases.